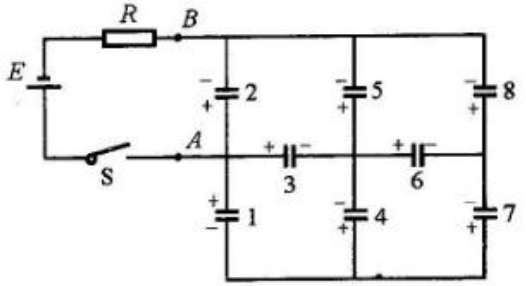
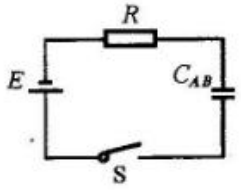
A、B 8 1

C_{AB} 2 C_{AB}



1



2

S A、B A、B

Q A B i q_i

1

4

$$q_1 + q_2 + q_3 = Q \tag{1}$$

$$-q_1 + q_4 + q_7 = 0 \tag{2}$$

$$-q_3 - q_4 + q_5 + q_6 = 0 \tag{3}$$

$$-q_6 - q_7 + q_8 = 0 \tag{4}$$

$$\frac{q_1}{C} + \frac{q_4}{C} - \frac{q_3}{C} = 0$$

$$q_1 + q_4 - q_3 = 0 \tag{5}$$

$$q_4 + q_6 - q_7 = 0 \tag{6}$$

$$q_2 - q_3 - q_5 = 0 \tag{7}$$

$$q_5 - q_6 - q_8 = 0 \tag{8}$$

2 C_{AB} E

(1) ~ (8)

(9)、(10)、(11)

$$E = \frac{q_2}{C} \tag{9}$$

$$E = \frac{Q}{C_{AB}} \tag{10}$$

$$q_2 = \frac{8}{15}Q \tag{11}$$

$$C_{AB} = \frac{15}{8}C \tag{12}$$

2 Q

$$W_0 = EQ \tag{13}$$

$$W_1 = \frac{1}{2}QE \tag{14}$$

$$W_2 = W_0 - W_1 \tag{15}$$